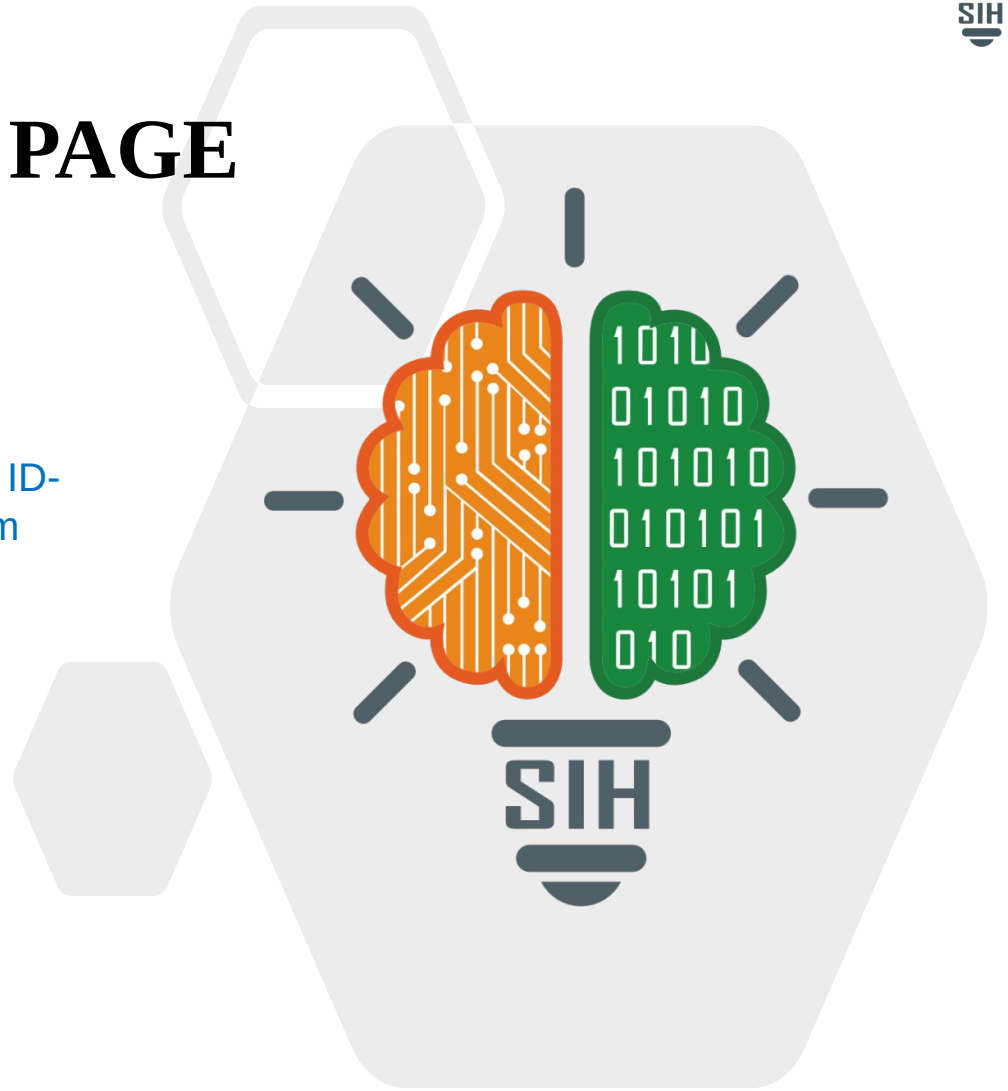
TITLE PAGE

- **Problem Statement ID –** SIH26263
- **Problem Statement Title-** Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** (as allotted on the SIH portal)
- **Team Name (Registered on portal)** AI Riders



AI Riders

VIDYA PRAMAAN

APAAR-linked student data, validation & verification platform for AICTE

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Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution



How it addresses the problem

✖ Only totals reported (except PGDM)	✔ Student-level records keyed on APAAR
✖ Excess admissions found late	✔ Seat #61 in a 60-seat course blocked at upload
✖ Same student at 2 colleges unnoticed	✔ Cross-college match: exact + fuzzy
✖ Records can't be verified at source	✔ Signed QR + consent-based check

Innovation and uniqueness of the solution

- **Explainable flags** — rule code + evidence
- **Live seat ledger** — intake & quota caps
- **Fuzzy identity match** — same person, new APAAR
- **ML risk score** — with plain reasons
- **Consent-first sharing** — student approves by OTP
- **Tamper-evident audit** — hash-chained log

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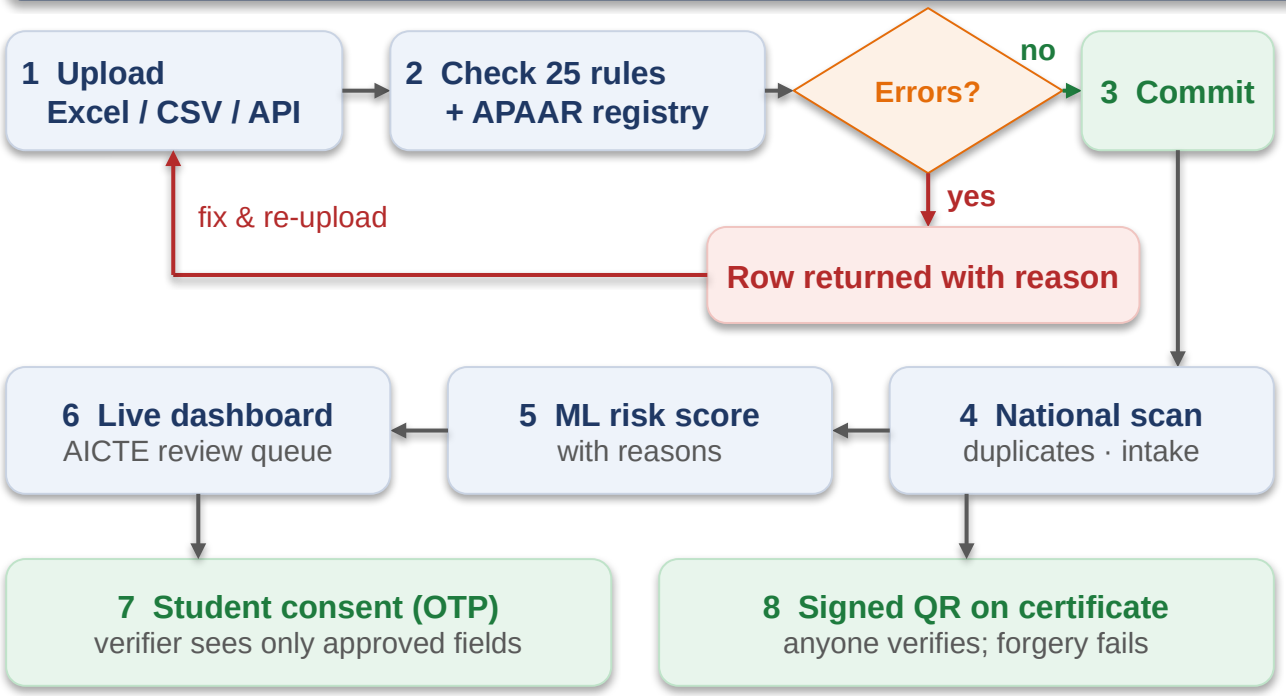
TECHNICAL APPROACH

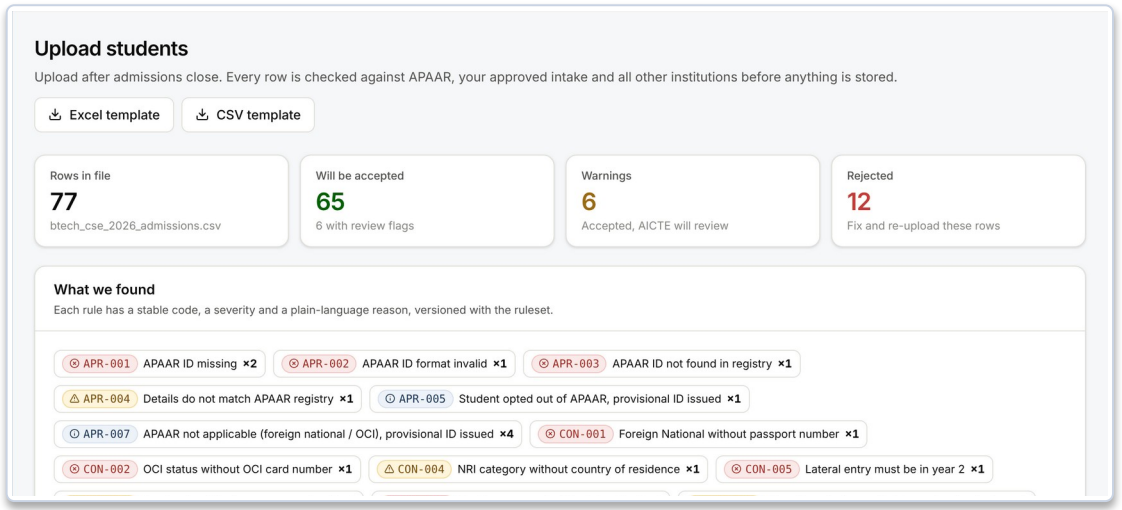
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❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Python · FastAPI** — REST + OpenAPI
- **React · TypeScript** — Tailwind · Recharts
- **SQLite → Postgres** — demo → production
- **scikit-learn** — Isolation Forest
- **RapidFuzz** — fuzzy matching
- **Ed25519 · W3C VC** — signed QR records

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)





Working prototype: 77-row upload → 65 accepted, 12 returned with rule codes

Run locally: ./run.sh
Demo users for every role · synthetic data

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

16/16
planted error types caught at upload

181/181
dual enrolments found across colleges

60/60
same person, different APAAR found

41/42
top-ranked colleges truly anomalous

0
false flags on 56 clean rows

~1 s
national scan of 48,053 enrolments

Measured by scripts/benchmark.py on 48,053 synthetic enrolments with planted anomalies · 25 automated API tests (SQLite + PostgreSQL) · browser walkthrough of all 5 roles

Potential challenges and risks	Strategies for overcoming these challenges
No public APAAR API	✔ Registry adapter: mock today → DigiLocker / API Setu after partner onboarding
Students without APAAR (opt-out, FN / OCI)	✔ Provisional ID: noted, never rejected
Student privacy (DPDP Act 2023)	✔ Consent per request, masked IDs, access log for the student
Wrong flags could hurt colleges	✔ Flags are review items; an officer decides; every action audited
Low digital readiness of small colleges	✔ Excel template with drop-downs, plain errors, Hindi toggle, mobile

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IMPACT AND BENEFITS

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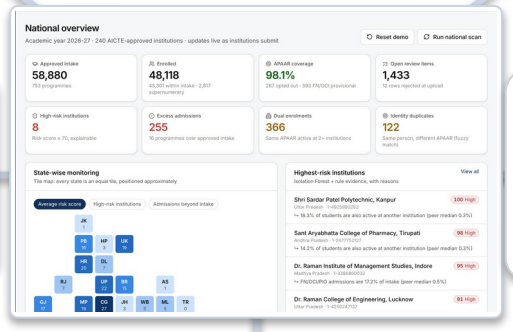
Potential impact on the target audience

AICTE & regulators
excess & dual enrolment caught at submission

Institutions
one upload, errors explained row by row

Policy makers
state / gender / category analytics & reports

Employers
instant QR check, consent-based facts



One national view

Students
lifelong record + control over who sees it

Benefits of the solution (social, economic, environmental, etc.)

Social
fair admissions; students own and control their record

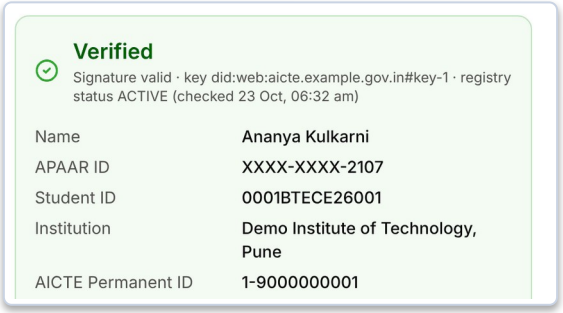
Economic
open-source, no licence cost; less manual scrutiny & fraud

Environmental
paperless returns and verification; no courier checks

Governance
real-time evidence for approvals & show-cause


Demo Institute of Technology, Pune
CERTIFICATE OF ENROLMENT
This is to certify that
Ananya Kulkarni
is a bona fide student of B.Tech Electronics & Communication (admitted 2026, year 1).
APAAR ID
6036 8720 2107
Student ID (Roll No.)
0001BTECE26001

Certificate: APAAR ID + student ID + QR


Verified
Signature valid · key did:web:aicte.example.gov.in#key-1 · registry status ACTIVE (checked 23 Oct, 06:32 am)

Name	Ananya Kulkarni
APAAR ID	XXXX-XXXX-2107
Student ID	0001BTECE26001
Institution	Demo Institute of Technology, Pune
AICTE Permanent ID	1-9000000001

QR scan: signature verified

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RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

What our research changed in the design

APAAR is created through Aadhaar-based verification	→ FN / OCI students get a provisional ID (APR-007)
APAAR consent is voluntary	→ Opt-out recorded, never rejected (APR-005)
DPDP Rules 2025 phase in to 2027	→ Consent, purpose limits, student access log built in
AICTE asks for APAAR-linked records on NAD / ABC	→ Feed NAD / ABC, not a parallel silo
APH sets supernumerary quotas	→ Quota caps are configuration, versioned

References

1. AICTE PS SIH26263, SIH 2026 portal sih.gov.in
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3. APAAR / Academic Bank of Credits abc.gov.in · digilocker.gov.in
4. DigiLocker Requester & Issuer APIs apisetu.gov.in
5. Digital Personal Data Protection Act 2023; DPDP Rules 2025 meity.gov.in
6. W3C Verifiable Credentials Data Model; RFC 8032 (Ed25519) w3.org/TR/vc-data-model · rfc-editor.org
7. Liu, Ting & Zhou (2008) Isolation Forest, IEEE ICDM doi.org/10.1109/ICDM.2008.17
8. Fellegi & Sunter (1969) A Theory for Record Linkage, JASA doi.org/10.1080/01621459.1969.10501049

Source code & tests
https://github.com/Pulkitkh/AI_Riders_
25 API tests + browser walkthrough